WAI ARIA

Alexander Thalhammer | @LX_T

WAI ARIA

- Web Accessibility Initiative
- Accessible Rich Internet Applications
- WAI ARIA
  - Roles
  - Attributes

Agenda

WAI ARIA

- Adds HTML-attributes to elements
  - improving accessibility for dynamic web content
  - especially for people with disabilities
- The attributes help assistive technologies
  - like screen readers, Braille, …
  - understand and navigate web content more effectively
- Defining roles, states, properties
  - relationships of elements
  - allowing Devs to create accessible & inclusive web apps

WAI-ARIA Basics

- Roles
  - enhance a11y by providing semantic information, when native HTML elements cannot be used
  - e.g. <div role="button" tabindex="0" (click)="onClick()">Click Me

- Attributes
  - enhance by providing additional semantic information
  - states (dynamic / temporal) vs. properties (permanent)

ARIA Roles I

- define what an element is or does
  - role="button" (prefer <button>)
  - role="navigation" (prefer <nav>)
  - role="complementary" (prefer <aside>)
  - role="main" (prefer <main>)
  - role="form" (prefer <form>)
  - role="banner"
  - role="search"
  - role="tablist" & role="tabpanel"
  - role="presentation"
    - removes default behaviour e.g. <button>

ARIA Roles II

- full list at
  - https://developer.mozilla.org/en-US/docs/Web/Accessibility/ARIA/Roles

ARIA Attributes I

- ARIA attributes enable modifying an element's states and properties as defined in the A11y tree

- Widget attributes
  - aria-disabled (state)
  - aria-expanded (state)
  - aria-hidden (state)
  - aria-required (property)

- Live region attributes (states)
  - aria-live (error message)
  - aria-busy (loading spinner)

ARIA Attributes II

- Drag-and-Drop attributes
  - aria-grabbed (state)
- Relationship attributes (properties)
  - aria-describedby
  - aria-details
  - aria-labelledby
- full list at
  - https://developer.mozilla.org/en-US/docs/Web/Accessibility/ARIA/Attributes

NG ARIA [attr]

- ARIA attributes enable modifying
  - an element's states and properties as defined in the A11y tree
  - https://developer.mozilla.org/en-US/docs/Web/Accessibility/ARIA/Attributes

- When binding to ARIA attributes in Angular, you must use the "attr." prefix

```
<!-- Use attr. when binding to an ARIA attribute -->
<button [attr.aria-label]="myActionLabel">…</button>
```

```
<!-- Static ARIA attributes require no extra -->
<button aria-label="Close">…</button>
```

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Questions?

ANGULAR
ARCHITECTS